

ServiceNow Certified System Administrator (CSA) Prep

Study Guide with Detailed Explanations

This comprehensive study guide provides detailed explanations for all key areas and topics required to pass the ServiceNow Certified System Administrator (CSA) exam. Each section includes core concepts, practical applications, and exam-focused insights.

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1. Service Catalog Management

Core Components

The Service Catalog is the central hub for users to request services and items. Understanding its components is crucial for the CSA exam.

Catalog Items

- Definition: Individual services or products that users can order from the catalog
- Purpose: Provide standardized way to request IT services, hardware, software, or other business services
- Key Features:
 - Variables for user input (text, choice lists, reference fields)
 - Approval workflows
 - Fulfillment processes
 - User criteria to control access
- Exam Focus: Know how to create catalog items, configure variables, and set user criteria

Record Producers

- Definition: Special type of catalog item that creates records in specific tables (not sc_req_item)
- Purpose: Allow users to create incidents, change requests, or other task records directly from the catalog
- Key Differences from Catalog Items:
 - Creates records in target tables (incident, change_request, etc.)
 - No approval process by default
 - Immediate record creation upon submission
- Common Examples: 'Create Incident', 'Password Reset', 'Report Outage'
- Exam Tip: Remember that Record Producers bypass the normal REQ/RITM process

Order Guides

- Definition: Collections of related catalog items that can be ordered together
- Purpose: Simplify ordering of multiple related items (e.g., new employee setup)
- Benefits:
 - Single request (REQ) for multiple items
 - Shared variables across items
 - Streamlined approval process
- Exam Focus: Understand when to use Order Guides vs individual Catalog Items

Service Catalog Variables

- Definition: Form fields that collect user input for catalog items
- Key Types:
 - Single Line Text: Basic text input
 - Multiple Choice: Radio buttons
 - Select Box: Dropdown list
 - Checkbox: True/false selection
 - Reference: Lookup to another table

- Date/Time: Calendar picker
- Important Properties:
 - Order: Controls display sequence on form
 - Mandatory: Required field validation
 - Read Only: Display-only field
- Variable Sets: Reusable collections of variables
- Exam Tip: Variables are global by default and can affect pricing

Request Fulfillment Process

- Request (REQ): The overall order placed by user
- Requested Item (RITM): Individual items within the request
- Catalog Task (SCTASK): Work tasks created for fulfillment
- Process Flow: REQ → RITM → SCTASK
- Exam Focus: Understand the relationship and navigation between these records

2. Data Management & Import Sets

Import Sets Overview

Import Sets provide a staging area for importing data from external sources into ServiceNow tables. This is a critical topic for data migration and integration scenarios.

Import Process Steps

1. Load Data: Import data from source (CSV, Excel, JDBC, LDAP) into Import Set table
2. Create Transform Map: Define field mappings between Import Set and target table
3. Transform Data: Execute the transformation to move data to target table
4. Clean Up: Remove processed data from Import Set table

- Exam Tip: Remember the exact sequence - Load, Map, Transform, Clean

Transform Maps

- Definition: Rules that define how data moves from Import Set table to target table
- Key Components:
 - Source table (Import Set table)
 - Target table (destination table)
 - Field maps (source field → target field)
 - Transform scripts (data manipulation)
- Mapping Options:
 - Automatic Mapping: When field names match exactly
 - Mapping Assist: Suggests mappings based on field names
 - Manual Mapping: Custom field-to-field mapping
- Reusability: Same Transform Map can be used multiple times

Coalesce Fields

- Definition: Fields used to identify existing records during import
- Purpose: Determine whether to update existing record or create new one
- Behavior:
 - Match found: Update existing record
 - No match found: Create new record
 - No coalesce fields: All records treated as new
- Best Practice: Use unique identifiers (email, employee ID, etc.)
- Exam Focus: Understand coalesce behavior and when updates vs inserts occur

Data Sources

- File Sources: CSV, Excel, XML files
- Database Sources: JDBC connections to external databases
- Directory Sources: LDAP/Active Directory
- Web Services: REST/SOAP APIs
- Exam Tip: Know which data sources are available and their use cases

3. User Interface & Navigation

Application Navigator

The Application Navigator is the primary navigation tool in ServiceNow, providing access to applications, modules, and functionality.

Navigation Components

- Applications: Top-level containers (e.g., Incident Management, Service Catalog)
- Modules: Specific functions within applications (e.g., Open Incidents, Create New)
- Favorites: Frequently accessed items marked with star icon
- History: Recently accessed items shown with clock icon
- Search: Filter navigator content by typing keywords
- Exam Focus: Know how to navigate and customize the navigator

Lists and Forms

- Lists: Display multiple records in tabular format
 - Column headers for sorting and filtering
 - Context menus for actions
 - Personalization options
 - List Editor for bulk updates
- Forms: Display single record details
 - Form header with record information
 - Form sections and fields
 - Related lists showing connected records
 - Form Designer for customization
- Exam Tip: Understand difference between Insert (create new) and Update (modify existing)

Filters and Search

- Filter Components: Field, Operator, Value
- Filter Types:
 - Condition Builder: Structured filtering
 - Breadcrumb: Visual filter display
 - Column Search: Quick field-specific filtering
 - Natural Language Query: Plain English filtering
- Global Search: Banner search for system-wide results
- Exam Focus: Know how to create and save filters

UI Customization

- Personalization: User-specific customizations
 - Column layout and order
 - Filter preferences
 - Form field arrangement
- System Customization: Admin-level changes
 - Form layout and design
 - List configurations
 - UI Actions (buttons, links, context menus)
- Basic Configuration UI16: Banner, colors, themes
- Exam Tip: Distinguish between user personalization and system configuration

4. Access Control & Security

Access Control Rules (ACLs)

Access Control Rules are the foundation of ServiceNow security, controlling who can access what data and perform which operations.

ACL Components

- Object: What is being secured (table, field, or record)
- Operation: What action is being controlled (create, read, write, delete)
- Permissions: Who has access (roles, scripts, conditions)
- ACL Types:
 - Table ACLs: Control access to entire tables
 - Field ACLs: Control access to specific fields
 - Record ACLs: Control access to specific records
- Naming Convention: table.field (e.g., incident.major_incident)

ACL Evaluation Order

1. Table-level rules (most specific to most general)
2. Field-level rules (most specific to most general)
3. Record-level rules (if applicable)

- Rule Specificity Order:

- Specific table name (incident)
- Parent table name (task)
- Wildcard (*)
- Exam Focus: Understand evaluation order and rule precedence

Roles and Groups

- Roles: Collections of permissions
 - Stored in sys_user_role table
 - Examples: admin, itil, catalog, approver_user
 - Can inherit from other roles
- Groups: Collections of users
 - Stored in sys_user_group table
 - Used for assignment and access control
 - Can contain roles
- Best Practices:
 - Assign roles to groups, not individual users
 - Add users to groups
 - Use principle of least privilege
- Exam Tip: Know the difference between roles and groups

User Management

- User Records: Stored in sys_user table
- User Authentication Methods:
 - Local database
 - LDAP/Active Directory
 - Single Sign-On (SSO)
- User Impersonation:
 - Testing and troubleshooting tool
 - Allows admins to see system as another user
 - Can be logged with glide.sys.log_impersonation property
- Exam Focus: Understand user authentication options and impersonation

5. CMDB & Configuration Management

CMDB Overview

The Configuration Management Database (CMDB) is a centralized repository that stores information about Configuration Items (CIs) and their relationships.

Key CMDB Tables

- cmdb_ci: Base table for all Configuration Items
- cmdb_rel_ci: Stores relationships between CIs
- cmdb: Contains CMDB-specific configuration
- Extended Tables: Specific CI types (cmdb_ci_server, cmdb_ci_computer, etc.)
- Exam Focus: Know the three core CMDB tables and their purposes

Configuration Items (CIs)

- Definition: Any component that needs to be managed to deliver IT services
- Types:
 - Hardware: Servers, computers, network devices
 - Software: Applications, databases, operating systems
 - Services: Business services, technical services
 - Documentation: Procedures, policies
- CI Attributes:
 - Operational status (Operational, Non-Operational, etc.)
 - Install status (Installed, In Stock, etc.)
 - Business criticality
- Exam Tip: Understand CI lifecycle and status values

CI Relationships

- Relationship Types:
 - Depends on: Service depends on server
 - Contains: Server contains hard drive
 - Connects to: Server connects to network
 - Runs on: Application runs on server
- Dependency View: Graphical representation of CI relationships
- Schema Map: Visual display of table relationships
- Exam Focus: Know common relationship types and visualization tools

CMDB Population Methods

- Discovery: Automated scanning of network infrastructure
- Service Mapping: Application dependency mapping
- Manual Entry: Direct data entry by users
- Import Sets: Bulk data import from external sources
- Integration Hub: Third-party system integration
- Exam Tip: Know which methods can populate the CMDB

Common Service Data Model (CSDM)

- Definition: ServiceNow's framework for organizing CMDB data
- Purpose: Standardize service-related definitions across platform
- Benefits:
 - Consistent data model
 - Better service reporting
 - Improved configuration strategies
- Exam Focus: Understand CSDM purpose and benefits

6. Workflow & Automation (Flow Designer)

Flow Designer Overview

Flow Designer is ServiceNow's modern automation platform that enables business process automation with minimal coding required.

Flow Components

- Triggers: Initiate flow execution
 - Record-based: When records are created/updated
 - Schedule-based: Time-driven execution
 - Application-based: External system events
- Actions: Perform work within the flow
 - Core Actions: Create Record, Look Up Record, Ask for Approval
 - Spoke Actions: Third-party integrations
 - Custom Actions: User-defined actions
- Data Pills: Runtime values passed between actions
- Conditions: Control flow execution based on criteria
- Exam Focus: Understand trigger types and core actions

Common Flow Use Cases

- Approval Processes: Multi-level approvals for requests
- Notifications: Automated email/SMS alerts
- Record Creation: Auto-generate related records
- Data Synchronization: Keep systems in sync
- Integration: Connect with external systems
- Exam Tip: Know when to use Flow Designer vs Business Rules

Integration Hub

- Definition: Platform for third-party system integration
- Spokes: Pre-built integrations for common systems
- Benefits:
 - No-code/low-code integration
 - Reusable components
 - Error handling and logging
- Exam Focus: Understand spoke concept and integration capabilities

7. Knowledge Management

Knowledge Base Structure

Knowledge Management provides a centralized repository for organizational knowledge, enabling users to find answers and share information efficiently.

Knowledge Articles

- Article States:
 - Draft: Being created/edited
 - Review: Under review process
 - Published: Available to users
 - Retired: No longer active
- Article Workflow: Controls publishing and retirement process
- Article Content: Text, images, attachments, links
- Exam Focus: Only Published articles are visible to end users

Knowledge Base Access Control

- User Criteria: Controls who can access knowledge base
- Access Tabs:
 - Can Read: Who can view articles
 - Can Contribute: Who can create/edit articles
 - Cannot Contribute: Exclusion list
- Public Knowledge: Accessible without login
- Default Access: All active users if no criteria specified
- Exam Tip: Understand different access levels and criteria

Knowledge Roles

- knowledge_admin: Manage multiple knowledge bases
- knowledge_manager: Manage specific knowledge base
- knowledge: Basic knowledge access
- Exam Focus: Know which roles can perform knowledge management tasks

Knowledge Search and Categories

- Search Results Sorting:
 - Relevancy: Best content match
 - Most recent update: Latest changes
 - Number of views: Popularity
- Categories: Organize articles by topic
- Tags: Additional classification and search
- Exam Tip: Know search sorting options and organization methods

8. Reporting & Analytics

Reporting Basics

ServiceNow provides comprehensive reporting capabilities for data analysis and visualization.

Report Creation

- Report Sources:
 - List column headers: Quick report from list view
 - Reports > Create New: Full report designer

- Context menus: Right-click options
- Report Types:
 - List: Tabular data display
 - Bar/Column Charts: Comparative data
 - Pie Charts: Proportional data
 - Line Charts: Trend analysis
- Exam Focus: Know multiple ways to create reports

Report Designer Sections

- Data: Specify table, fields, and filters
- Type: Choose report visualization type
- Configure: Set grouping and calculations
- Style: Customize appearance and formatting
- Exam Tip: Understand purpose of each designer section

Report Sharing and Access

- Report Visibility:
 - My Reports: Personal reports
 - Group: Shared with specific groups
 - Global: Available to all users
 - All: System-wide visibility
- Dashboards: Collections of reports and widgets
- Scheduled Reports: Automated report generation
- Exam Focus: Know report sharing options

Performance Analytics

- Definition: Advanced analytics with historical data snapshots
- Key Differences from Reporting:
 - Time-series data collection
 - Trend analysis capabilities
 - Forecasting functionality
 - KPI tracking and monitoring
- Use Cases: Performance monitoring, trend analysis, capacity planning
- Exam Tip: Understand when to use Performance Analytics vs standard reporting

Metrics

- Definition: Measurements used to evaluate IT service management effectiveness
- Purpose: Track performance, identify improvements, support decision-making
- Display: Homepage gauges, dashboards, reports
- Exam Focus: Know metric definition and purpose

9. Platform Administration

System Properties

System properties control various aspects of ServiceNow platform behavior and appearance.

Basic Configuration UI16

- Customizable Elements:
 - Banner image and colors
 - Browser tab title
 - System date format
 - Header background color
 - Base theme selection
- Access: System Properties > Basic Configuration UI16
- Exam Focus: Know what can be customized through basic configuration

High Security Settings

- Purpose: Enhanced security for sensitive operations
- Access Requirements:
 - security_admin role
 - Elevate Roles action
- Return to Normal: Log out and back in
- Exam Tip: Know how to access and exit high security mode

System Dictionary

- Definition: Contains definition for every table and column
- Information Stored:
 - Field properties and attributes
 - Data types and constraints
 - Display labels and help text
- Dictionary Overrides: Customize field properties in extended tables
- Exam Focus: Understand dictionary purpose and overrides

Plugins

- Definition: Add-on functionality for ServiceNow platform
- Purpose: Enable additional applications and features
- Activation: System Definition > Plugins
- Examples: HR Service Delivery, IT Operations Management
- Exam Tip: Know that plugins add applications/functionality

10. Business Rules & Client Scripts

Business Rules

Business Rules are server-side scripts that execute when database operations occur.

Business Rule Components

- Required Elements:
 - Table: Which table the rule applies to
 - When: Timing of execution (Before, After, Async, Display)
 - Condition: When the rule should run
 - Script: JavaScript code to execute
- Timing Options:
 - Before: Execute before database operation
 - After: Execute after database operation
 - Async: Execute asynchronously
 - Display: Execute when form loads
- Exam Focus: Know required components and timing options

Client Scripts

Client Scripts execute in the user's browser to enhance form functionality.

Client Script Types

- onLoad: Executes when form loads
- onChange: Executes when field value changes
- onSubmit: Executes when form is submitted
- onCellEdit: Executes when list cell is edited
- Key Differences from Business Rules:
 - Client Scripts: Run in browser (client-side)
 - Business Rules: Run on server (server-side)
- Exam Focus: Know client script types and client vs server execution

11. UI Policies & Data Policies

UI Policies

UI Policies control form behavior and field properties without requiring scripting.

UI Policy Capabilities

- Field Control:
 - Make fields mandatory
 - Make fields read-only
 - Hide/show fields
- Execution: Client-side (browser)
- Scope: Form interactions only
- Global Setting: Apply to all views when set to true
- Exam Focus: Know UI Policy capabilities and limitations

Data Policies

Data Policies enforce data consistency regardless of how data enters the system.

Data Policy Characteristics

- Execution: Server-side
- Scope: All data entry methods
 - Form submissions
 - Import Sets
 - Web Services
 - API calls
- Use Cases:
 - Data validation
 - Mandatory field enforcement
 - Read-only field protection
- Exam Tip: Data Policies run regardless of data entry method

UI Policy vs Data Policy

- UI Policies: Form interactions only, client-side
- Data Policies: All data entry, server-side
- When to Use:
 - UI Policy: Form-specific behavior
 - Data Policy: Data integrity across all entry points
- Exam Focus: Understand when to use each type

12. Update Sets & Application Management

Update Sets Overview

Update Sets capture customizations and move them between ServiceNow instances.

Update Set Process

- Development Instance:
 1. Create named Update Set
 2. Make customizations
 3. Mark Update Set as Complete
- Target Instance:
 4. Retrieve Update Set
 5. Preview Update Set
 6. Commit Update Set
- Exam Focus: Know the complete Update Set process

Update Set Contents

- Included by Default:
 - Tables, forms, views, fields
 - Business Rules, Client Scripts
 - UI Policies, UI Actions
 - Workflows, Flow Designer flows
- Not Included by Default:

- Data records
- Schedules
- Homepages/Dashboards
- Scheduled Jobs
- Manual Addition: Some items must be manually added
- Exam Tip: Know what's included and excluded by default

Update Set Best Practices

- Avoid Default Update Set: Don't use for moving customizations
- Named Update Sets: Create specific Update Sets for projects
- Batching: Group related Update Sets for easier management
- Testing: Always preview before committing
- Exam Focus: Know best practices and Default Update Set limitations

Application Scope

- Definition: Boundary that protects applications and restricts access
- Purpose: Application isolation and security
- Global Scope: Platform-wide access
- Custom Scope: Application-specific boundaries
- Exam Tip: Understand scope concept and security benefits

Exam Preparation Tips

Key Study Strategies

- Focus on Practical Application: Understand how features work together
- Hands-on Practice: Use a Personal Developer Instance (PDI)
- Memorize Key Facts: Table names, field names, process steps
- Understand Relationships: How different components interact
- Review Common Scenarios: Real-world use cases and solutions

Common Exam Traps

- Table Names: Know exact table names (sys_user, not user)
- Process Order: Remember exact sequence of steps
- Default Behaviors: Understand what happens by default
- Scope and Permissions: Know what requires specific roles
- Feature Limitations: Understand what each feature can/cannot do

Final Checklist

- ☐ Service Catalog components and request process
- ☐ Import Set process and Transform Maps
- ☐ Navigation and UI customization
- ☐ Access Control evaluation and security

- ☐ CMDB structure and CI relationships
- ☐ Flow Designer components and triggers
- ☐ Knowledge Management access and workflow
- ☐ Reporting creation and sharing
- ☐ Business Rules vs Client Scripts
- ☐ UI Policies vs Data Policies
- ☐ Update Set process and contents
- ☐ Platform administration basics

This study guide covers all essential topics for the ServiceNow CSA exam. Practice with hands-on exercises and review the official ServiceNow documentation for additional details. Good luck with your certification!